



The Effects of Digital Payments on the Financial Systems in India

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Introduction

Motivation: The Unified Payments Interface (UPI) catalyzed the most rapid digital financial adoption in modern history. However, a paradox emerged: despite frictionless microeconomic efficiency, physical Currency in Circulation (CIC) remains remarkably persistent.

Research Problem: Does the transition to real-time digital payments fundamentally alter the classical macroeconomic structure of the Indian economy?

Research Gaps: There is a severe lack of empirical evidence regarding how zero-cost digital transactions affect aggregate money velocity, institutional policy pass-through and liquidity preference in emerging economies.

Objectives: To empirically test for structural breaks in long-run money velocity, evaluate shifts in commercial bank policy pass-through, and measure changes in retail liquidity preference (the M1/M3 ratio).

Review of Literature

Awasthy et al. (2022): Formally identified India's "Currency Demand Paradox," documenting the simultaneous, counter-intuitive surge in both digital payment adoption and precautionary cash hoarding.

Sarkisyan (2024): Investigated Brazil's Pix system, providing the first major evidence on how zero-friction, real-time payment networks influence commercial banking behavior and institutional liquidity constraints.

Awasthy & Seet (2025): Examined the direct impact of UPI on aggregate cash demand at both national and subnational levels, highlighting the ongoing tension between rapid digital adoption and the persistence of physical currency.

Data and Methods

Data Timeframe: Quarterly macroeconomic datasets from 1996 Q2 to 2025 Q1, providing a robust, multi-decade pre-digital baseline. [Source RBI, NPCI]

Core Variables: GDP, Narrow Money (M1), Broad Money (M3), Logarithmic UPI Transaction Volume, RBI Repo Rate, and Weighted Average Lending Rates (WALR).

Structural Break Testing: Deployed the Zivot-Andrew's algorithm to isolate whether historical breaks were technological or exogenous.

Econometric Modeling: Utilized ARDL bounds testing for long-run equilibriums.

Diagnostic Correction: Deployed Newey-West standard errors to correct for diagnostically proven, severe serial correlation in short-run models.

$$(M_1/M_3)_t = \beta_0 + \beta_1 \ln(UPI_Volume_t) + \beta_2(RepoRate_t) + \epsilon_t$$

Key Findings

Velocity is Anchored: Data definitively rejects a long-run structural break. UPI volume shares no permanent cointegration with V1 or V3 velocity.

Shocks are Exogenous: The Zivot-Andrew's test pinpointed the 2020 pandemic as the primary break in aggregate velocity, proving recent deviations are non-technological anomalies.

Institutional Inertia: Digital transaction volume does not significantly accelerate the RBI's policy rate pass-through ($p = 0.266$).

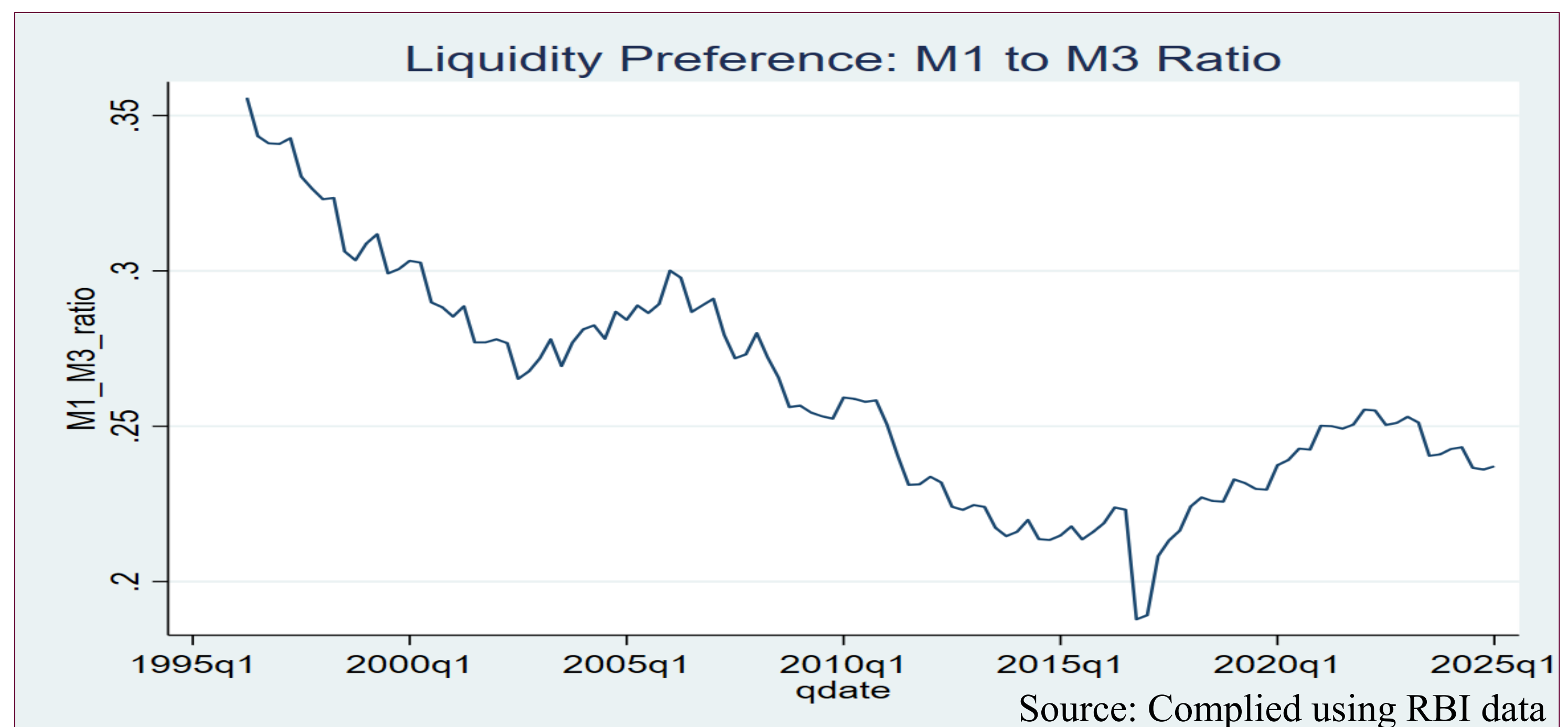
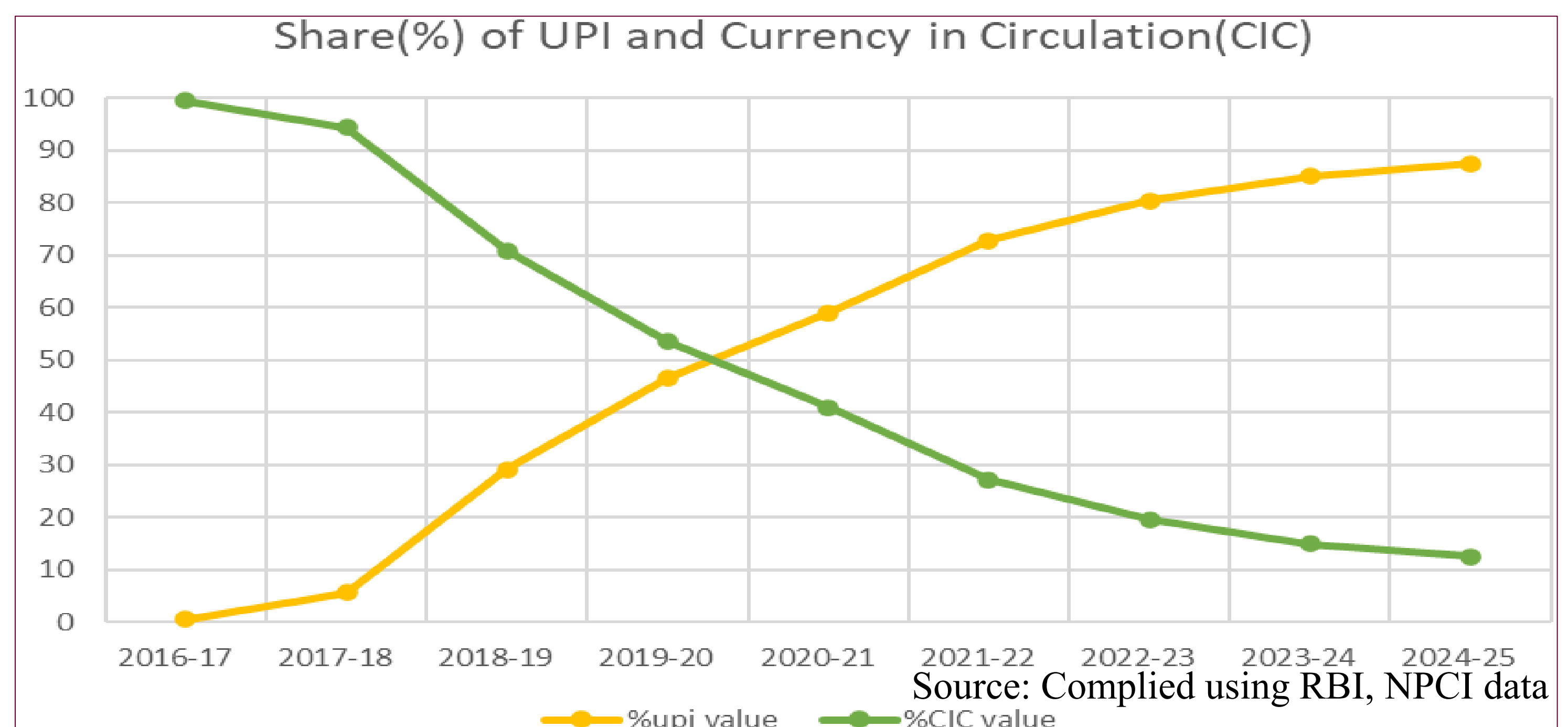
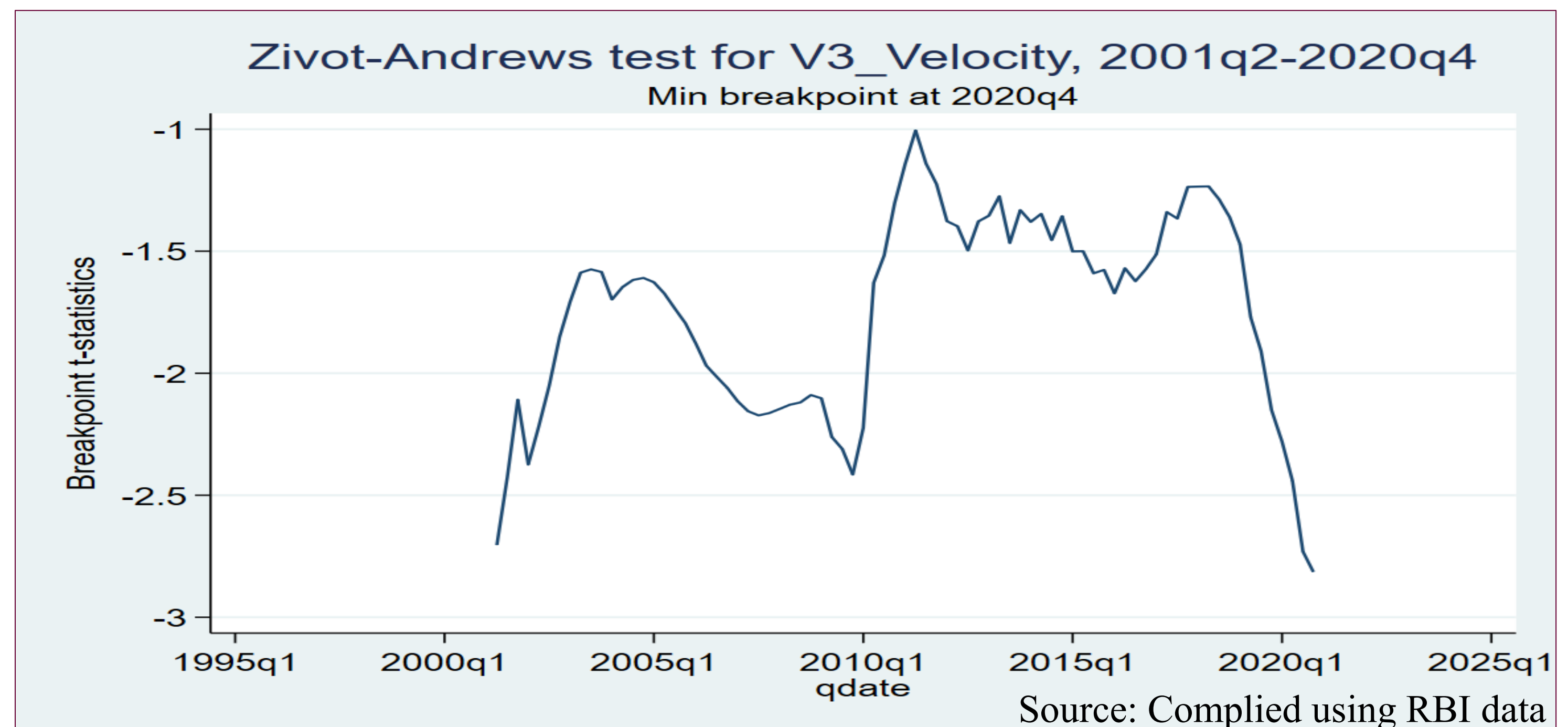
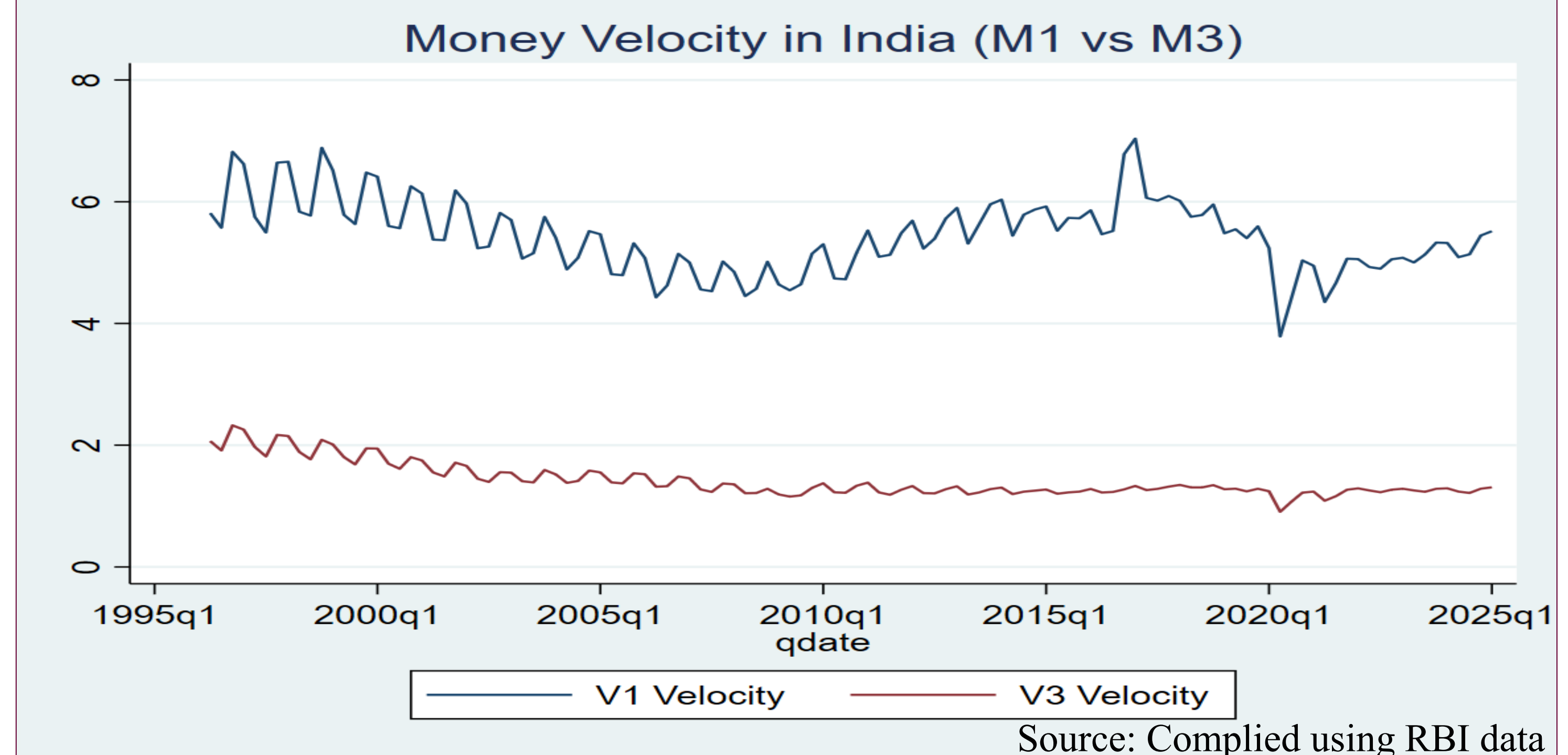
A Significant Micro-Shift: Newey-West regressions confirmed a statistically significant negative relationship ($p = 0.041$) between digital volume and the short-run M1/M3 composition ratio.

Discussion

The Modern Baumol-Tobin Effect: Because the digital network drives transaction costs to effectively zero, the traditional *transactions motive* for hoarding liquid cash has been neutralized.

Retail Optimization: Economic agents now actively minimize zero-yield M1 balances. They park surplus capital in higher-yielding time deposits (M3) and utilize the frictionless UPI network to transfer exact fractional amounts instantly at the point of sale.

The Paradox Resolved: India's financial digitization is a microeconomic triumph that perfected the medium of exchange without breaking the fundamental laws of classical macroeconomics.



Conclusion

The rapid adoption of digital payments in India has completely transformed retail liquidity management without destabilizing the broader macroeconomic architecture. While classical velocity and institutional pass-through remain stable, the micro-level optimization of wealth is undeniable. Moving forward, the RBI must recalibrate its high-frequency forecasting models; legacy systems targeting broad money will invariably over-predict the demand for transactional liquidity in a hyper-efficient, digitized retail tier.

References

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